

## **Stat 2300 International, Fall 2006 – Midterm Answers**

Friday, October 27, 2006

1. **b.** frame.
2. **c.** 75<sup>th</sup>
3. **d.** increases, decreases
4. **a.** the z distribution.
5. **a.** sampling with replacement.
6. **c.** the event occurring in one interval is independent of the event occurring in any other nonoverlapping interval, and the random variable of interest is discrete.
7. **a.** Undercoverage.
8. **a.** The median is resistant to extreme values.
9. **c.** larger, larger.
10. **c.** symmetric.
11. **a.** sample space.
12. **b.** Zero
13. **a.**  $\sum x p(x)$
14. **d.** None of the above.
15. **d.**  $(n)(p)(1-p)$
16. **d.** 20
17. **b.** has a below average price-to-earnings ratio.
18. **b.** increases, standard normal curve.
19. **b.** greater than
20. **d.** 48 and 1.

21.  $z = (21.5 - 20)/.6 = 2.5$ ; value from normal table = 0.4938;  
Probability =  $0.5 - 0.4938 = 0.0062 = 0.62\%$
22.  $.2 + .2 = .4 = 40.00\%$
23.  $[P(B) = .4, P(M) = .5, P(B|M) = .2]$   
 $P(M|B) = P(B \cap M) / P(B) = P(M) * P(B|M) / P(B) = .5 * .2 / .4 = .1 / .4 = .25 = 25\%$
24.  $P(6 \leq x \leq 6.5) = P((6-5)/.5 \leq z \leq (6.5-5)/.5) = P(2 \leq z \leq 3) =$   
 $= .9987 - .9772 = .0215 = 2.15\%$
25. (i) 4.6 (ii) 5 (iii) 7
26. for 99%CI:  $z_{\alpha/2} = 2.575$ ;  
CI:  $[\bar{x} \pm z_{\alpha/2} * \sqrt{s^2/n}] = [.4 \pm 2.575 * .04924] = [.2732, .5268]$
27. [mean = 20] (i) 10 (ii) 10.4 (iii) about 3.22
28. Probability =  $(1-.95)*(1-.90) = .05*.10 = 0.005 = 0.5\%$
29. Expected Profit =  $0.005*(-49,600) + 0.995*400 = \$150$
30.  $\mu = np = 400*.4 = 160$ ,  $\sigma = \sqrt{np(1-p)} = \sqrt{400*.4*.6} =$   
 $= \sqrt{96} = 9.7980$   
 $P(x = 170) = P(169.5 \leq x \leq 170.5) = P((169.5-160)/9.7980 \leq z \leq (170.5-160)/9.7980) = P(.96 \leq z \leq 1.07) = .8577 - .8315 = .0262 = 2.62\%$
31.  $P(x = 0) = e^{(-2.5)} * 2.5^0 / 0! = .0821 = 8.21\%$
32.  $\bar{x} = \text{mean} = 50$ ,  $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 1 / \sqrt{144} = .0833$ ;  
 $P(\bar{x} > 50.2) = P(z > (50.2-50)/.0833) = P(z > 2.40) = 1 - .9918 = .0082 = 0.82\%$
33. Probability =  $12! / (11! * 1!) * .8^{11} * .2^1 = .2062 = 20.62\%$
34. Mean 31.55306122  
Confidence Level(98.0%) 0.27477828  
98% CI: 31.27828294 31.8278395
35. (i)  
Mean 159.7692308  
(ii)  
Median 152  
(iii)  
[Sample Variance 939.025641] =>  
Population Variance =  $12/13 * 939.025641 = 866.7928994$